

MATA37H3 — Week 02

Answer Key & Misconception Notes

For tutor use only — do not show the student until after their attempt. (The Fundamental Theorem of Calculus)

EXAMPLE 1 — Answer

$$\frac{3x^9}{9} - \frac{18x^6}{6} + x + C = \frac{x^9}{3} - 3x^6 + x + C; \int \sqrt[3]{x} dx = \int x^{1/3} dx = \frac{3}{4}x^{4/3} + C$$

Common misconception: Using the power rule on $x^{1/3}$ but writing the new exponent as $1/3 - 1 = -2/3$ without flipping the sign correctly, or forgetting to invert-and-multiply by the reciprocal of the new exponent ($\frac{3}{4}$, not $\frac{4}{3}$ inverted incorrectly).

EXAMPLE 2 — Answer

$$\int \sec x \tan x dx = \sec x + C; \int \frac{1}{x+5} dx = \ln|x+5| + C$$

Common misconception: Confusing $\sec x \tan x$'s antiderivative with $\tan x$'s (writing $\tan x + C$ instead of $\sec x + C$) — these are two different “guess and check” facts that get mixed up.

EXAMPLE 3 — Answer

Domain $x > 0$ (since $3x > 0 \iff x > 0$); $F'(x) = \frac{1}{3x} \cdot 3 = \frac{1}{x}$; matches $\ln x + C$ with $C = \ln 3$, since $\ln(3x) = \ln 3 + \ln x$

Common misconception: Computing $\frac{d}{dx} \ln(3x)$ as $\frac{1}{3}$ (forgetting the chain rule multiplies by the derivative of the *inside*, which is 3, and the 3s cancel to leave exactly $1/x$, not $1/(3x)$ or $1/3$).

EXAMPLE 4 — Answer

(a) valid; (b) **not** valid — the free variable x appears both as the upper limit and inside the integrand, which is not allowed; (c) **not** valid for two separate reasons: the limits are backwards-oriented as written to look like an antiderivative in x , and more importantly x appears in the integrand while the integral is not with respect to x — this is fine on its own (that's what makes it a function of x), so actually flag (c) as valid **only if** read as $F(x) = \int_3^0 (\dots) dt$, a legitimate (if unusually oriented) function of x

Common misconception: Assuming that *any* appearance of the outer variable x inside the integrand invalidates the definition — it doesn't, as long as x is not *also* claiming to be the dummy variable of integration (that's specifically case (b)'s problem).

EXAMPLE 5 — Answer

$$F'(x) = \cos(x^2)$$

Common misconception: Adding an extra chain-rule factor when the upper limit is literally x (not a function of x) — no extra multiplication is needed here, unlike Problems 6–7.

EXAMPLE 6 — Answer

$F'(x) = -\sin^3(\sqrt{x}) \cdot \frac{1}{2\sqrt{x}}$, i.e. $F'(x) = -\sin^3(\sqrt{x})$, using $F(x) = -\int_7^x \sin^3(\sqrt{t}) dt$ first to flip the limits, then differentiating (the “inside function” here is just x itself, so no extra chain-rule factor beyond the sign flip)

Common misconception: Forgetting to flip the sign when the variable is in the *lower* limit, and reporting $F'(x) = \sin^3(\sqrt{x})$ without the minus sign.

EXAMPLE 7 — Answer

$$F'(x) = \frac{e^x}{1+e^{8x}} \cdot e^x - \frac{\sin x}{1+\sin^8 x} \cdot \cos x$$

Common misconception: Only differentiating with respect to the upper limit and forgetting the lower limit is *also* a function of x here — both limits need their own chain-rule term, with a minus sign on the lower-limit term.

EXAMPLE 8 — Answer

- (a) False — G could be $\int_0^x f(t) dt + 5$ for instance, which is also an antiderivative but isn’t equal to the bare integral unless $G(0) = 0$; (b) True — this is exactly the “ $+C$ ” fact, since both $G(x)$ and $\int_0^x f(t) dt$ are antiderivatives of f on the single interval $\mathbb{R}$

Common misconception: Believing that every antiderivative must be expressible with lower limit exactly 0 — the constant C can be realized using $\int_0^x f dt$ with any added constant, or equally well using $\int_{x_0}^x f dt$ for any other base point x_0 ; there’s no canonical choice.

EXAMPLE 9 — Answer

$$[e^x - e^{-x} - \frac{1}{2} \sin(2x)]_0^1 = (e - e^{-1} - \frac{1}{2} \sin 2) - (1 - 1 - 0) = e - e^{-1} - \frac{1}{2} \sin 2$$

Common misconception: Mis-antidifferentiating e^{-x} as e^{-x} instead of $-e^{-x}$ (sign error from the chain rule on the exponent $-x$), which flips the sign of that whole term in the final answer.

EXAMPLE 10 — Answer

$$[\tan x]_{\pi/4}^{\pi/3} = \tan(\pi/3) - \tan(\pi/4) = \sqrt{3} - 1$$

Common misconception: Writing the antiderivative of $\sec^2 x$ as $\sec x \tan x$ (confusing it with the derivative of $\sec x$) instead of $\tan x$.

EXAMPLE 11 — Answer

Curves meet where $x^2 + 3 = 3x + 1 \Rightarrow x^2 - 3x + 2 = 0 \Rightarrow x = 1, 2$; on $[1, 2]$ the line is on top, so Area = $\int_1^2 [(3x + 1) - (x^2 + 3)] dx = \int_1^2 (-x^2 + 3x - 2) dx = \left[-\frac{x^3}{3} + \frac{3x^2}{2} - 2x\right]_1^2 = \frac{1}{6}$

Common misconception: Integrating $(x^2 + 3) - (3x + 1)$ (top minus bottom reversed) without first checking which curve is actually larger on $[1, 2]$, producing a negative “area.”

EXAMPLE 12 — Answer

Average = $\frac{1}{3-0} \int_0^3 (9 - x^2) dx = \frac{1}{3} \left[9x - \frac{x^3}{3}\right]_0^3 = \frac{1}{3}(27 - 9) = 6$; solve $9 - c^2 = 6 \Rightarrow c^2 = 3 \Rightarrow c = \sqrt{3}$ (taking the root in $[0, 3]$)

Common misconception: Forgetting to divide by $b - a$ when computing the average value — reporting the bare integral 18 as if it were the average, instead of $18/3 = 6$.